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To cite this article: Weihua Dong, Yulin Wu, Tong Qin, Xinran Bian, Yan Zhao, Yanrou He, Yawei Xu & Cheng Yu (2021) What is the difference between augmented reality and 2D navigation electronic maps in pedestrian wayfinding?, *Cartography and Geographic Information Science*, 48:3, 225-240, DOI: [10.1080/15230406.2021.1871646](https://doi.org/10.1080/15230406.2021.1871646)

To link to this article: <https://doi.org/10.1080/15230406.2021.1871646>



Published online: 02 Feb 2021.



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ARTICLE



What is the difference between augmented reality and 2D navigation electronic maps in pedestrian wayfinding?

Weihua Dong ^a, Yulin Wu ^b, Tong Qin ^a, Xinran Bian ^a, Yan Zhao ^a, Yanrou He ^a, Yawei Xu ^a
and Cheng Yu^a

^aFaculty of Geographic Science, Beijing Normal University, China; ^bFaculty of Psychology, Beijing Normal University, China

ABSTRACT

Augmented reality (AR) navigation aids have become widely used in pedestrian navigation, yet few studies have verified their usability from the perspective of human spatial cognition, such as visual attention, cognitive processing, and spatial memory. We conducted an empirical study in which smartphone-based AR aids were compared with a common two-dimensional (2D) electronic map. We conducted eye-tracking wayfinding experiments, in which 73 participants used either a 2D electronic map or AR navigation aids. We statistically compared participants' wayfinding performance, visual attention, and route memory between two groups (AR and 2D map navigation aids). The results showed their wayfinding performance did not differ significantly. Regarding visual attention, the participants using AR tended to have significantly shorter fixation durations, greater saccade amplitudes, and smaller pupil sizes on average than the 2D map participants, which indicates lower average cognitive workloads throughout the wayfinding process. Considering attention on environmental objects, the participants using AR paid less visual attention to buildings but more to persons than the participants using 2D maps. Sketched routes results revealed that it was more difficult for AR participants to form a clear memory of the route. The aim of this study is to inspire more usability research on AR navigation.

ARTICLE HISTORY

Received 15 July 2020
Accepted 31 December 2020

KEYWORDS

Augmented reality; 2D electronic map; navigation aids; pedestrian navigation; eye-tracking; wayfinding; usability

1. Introduction

Navigation is defined as a goal-directed moving process toward a relatively clear destination in an environment (Montello, 2005). People need to obtain and process spatial information, such as location, direction, and distance, to make wayfinding decisions (Montello & Raubal, 2013). To reduce users' cognitive workload, a variety of external aids are used to help users to navigate more efficiently. In recent years, augmented reality (AR) technology has developed quickly and become widely applied in tasks in the field of geography, such as geovisualization (Hedley et al., 2002), geography education (Carrera et al., 2018), and the development of geographical information systems (Luchetti et al., 2017). In particular, AR navigation systems installed on mobile devices combine the video stream captured by a smartphone camera with positioning and orientation data flow (Amirian & Basiri, 2016). These developments demonstrate crucial technological progress from the original head-mounted form (Thomas et al., 1998). In contrast to traditional two-dimensional (2D) electronic maps that use abstract symbols to represent geographic entities, an AR system shows real scenes and landmarks on the smartphone screen, which expresses geographical information and relationships more specifically. Furthermore, AR

generates virtual graphics (often accompanied with text instructions) overlaid on the real scene (Zlatanova, 2002). Cognitive effort, such as mental rotation and landmark feature association, required when using a map, is eliminated. Users only need to understand the straightforward guidance shown on the interface and make corresponding spatial decisions without the need to interpret abstract symbols or manipulate internal spatial representations. Some researchers believe that AR facilitates cognitive activities during wayfinding and improves the efficiency of navigation (Seager & Fraser, 2007).

However, the emergence of AR aids has also raised many concerns. AR aids might negatively influence the spatial learning process and result in poor spatial memory. Some researchers have suggested that there might be a trade-off between the usability of such aids and the acquisition of spatial knowledge (Münzer et al., 2011). The higher the level of abstract representation, the more cognitive effort is required for interpretation, and this is often accompanied by a deeper level of spatial comprehension (Duke et al., 2003); that is, AR's usability might be achieved at the expense of users' spatial learning processes (Willis et al., 2009). However, empirical evidence to support such views is still lacking. Another

concern is whether AR user safety is threatened because of attentional alternation. Researchers have found that the use of virtual navigation aids could alter the visual attention of users, and even cause some safety problems, such as inattentive blindness (Bertrand et al., 2013; Dixon et al., 2013; Haines, 1991). An AR user's interactive behavior patterns might also differ from those of a 2D map user; however, there have been few studies on this topic.

The aim of our research is to compare the two aids introduced above, to determine their strengths and weaknesses, and to expand the evidence regarding the performance of AR aids through field experiment-based research. Specifically, we aim to answer the following questions:

- Will the adoption of AR aid improve wayfinding performance when compared with the use of a 2D electronic map?
- How are the characteristics and distribution of visual attention altered by an AR navigation aid? How do the average attention metrics and attentional allocation across environment objects differ between AR and 2D map users?
- Does an AR aid have any influence on spatial knowledge acquisition and storage when it is used instead of a 2D electronic map? If so, is it a positive or negative influence?

In this study, we designed a two-factor wayfinding experiment to compare the navigational efficiency and evaluate the usability of two navigation aids: a smartphone-based AR navigation system and a 2D electronic map. We recruited participants to complete several wayfinding tasks. In the experiment, we collected their behavioral data, eye-tracking data, and sketch maps (Section 3). After the experiment, we quantitatively analyzed the data, and several findings were revealed (Section 4). We discuss the differences between the two navigation aids based on statistical results (Section 5) and present the conclusions of this research (Section 6).

2. Related work

2.1. Role of AR and 2D map navigation aids in wayfinding

Before the emergence of AR navigation aids, 2D electronic maps were widely used for navigation. Previous studies have evaluated various aspects of electronic navigation aids, thereby accumulating a wealth of academic knowledge. For instance, on the topic of wayfinding, Münzer et al. (2006) indicated that users make fewer

mistakes and arrive at the destination more quickly using a 2D electronic map. On the topic of self-localization and orientation, Kiefer et al. (2013) explored how the matching process from landmarks to map symbols takes place when wayfinding. The research reported successful participants paid more attention to critical map symbols and made more switches between map and surroundings during self-localization and spatial orientation tasks. Regarding spatial knowledge acquisition, researchers found that navigation assistance leads to better knowledge acquisition, but poorer knowledge recollection (Krüger et al., 2004; Münzer et al., 2006). With respect to usability, previous studies concluded that 2D electronic maps can adjust their presentation according to changes in the users' position, thereby catering to their usage habits (Burnett & Joyner, 1996). On the topic of individual differences, different forms of visualization can strongly influence navigation performance based on demographic variables, such as gender (Liao & Dong, 2017; Zhou et al., 2020) or spatial ability (Rodes & Gugerty, 2012). Furthermore, earlier studies found that different navigation aids influence the acquisition of geographic information and popularity among users. Studies have shown that electronic navigation aids are more popular than paper maps among bicycle riders, and riders with poor spatial abilities benefit substantially from turning information, which can be presented in voice form (Waard et al., 2017). In terms of the research environment, laboratory experiments have been widely adopted because of their feasibility, controlled environment, and internal validity (Liao et al., 2016; Spiers & Maguire, 2008), but outdoor studies, with participants holding aids and navigating themselves in a real scene, have also been popular (Liao et al., 2019).

The development of AR technology has catalyzed the emergence of AR navigation aids. Some scholars have claimed that AR is a subclass of mixed reality. Virtual graphics are overlaid on the real scene images and are thus expected to enhance the effect of the display (Milgram & Kishino, 1994). It is reasonable to believe that the user experience of AR aids differs from that of 2D electronic maps as a result of their distinctive features. First, unlike 2D maps, AR aids do not require users to recognize abstract symbols and conduct complex map-learning tasks, such as locating themselves and route planning. The aids' virtual graphics are directly embedded into the real-time scene. They update and present the scene as the attitude of the device changes, and thus facilitate navigation using their vivid and immersive wayfinding instructions (Gardony et al., 2018; Liu & Meng, 2020). Second, AR aids typically adopt a track-up display rather than north-up display used by many types of map. When the orientation of

external aids is aligned with the direction in which users are moving, a turning decision is faster and more accurate (Viita & Werner, 2006). This can be because of discrepancies in the frame of reference adopted (Hofer & Wickens, 1997; Wickens & Preveit, 1995). Users of track-up navigation aids with an egocentric reference system make the local decisions of turning left or right more easily than users of north-up ones with an exocentric system, but the latter may gain better spatial knowledge (Aretz, 1991). Third, AR aids might have a higher level of automation than 2D maps, making some operations unnecessary, such as zoom in/out, pan, and rotation. Users only need to walk under the guidance of virtual graphics. Increasing the level of automation of navigation aids might alter navigation performance, gaze behavior, and the acquisition of spatial knowledge (Brügger et al., 2019). Revealing the differences in people's behavior, attention, and cognitive processes between AR aids and 2D maps is worth studying, considering that they have different display configurations and varying extents of automation.

Studies have focused on both the hardware and software aspects of AR navigation systems. Feiner et al. (1997) from Columbia University developed a prototype of an AR navigation system that was used in an urban environment, mainly for navigation on a university campus. Subsequently, more researchers have increasingly focused on the development and application of AR navigation aids in indoor environments, such as airports, museums, and libraries (Gerstweiler et al., 2015; Huang et al., 2016; Wang, 2019), in addition to outdoor environments. A few studies have evaluated the behavior of AR navigation users, mainly comparing AR systems with 2D electronic maps, and focusing on efficiency and user experience. Kluge and Asche (2012) described a pilot study on an AR navigation aid. They developed a smartphone-based navigation module, whose presentation mode could be easily switched between AR and a 2D map by adjusting the attitude of the phone. Using eye-tracking technology, they found that users tend to switch to 2D maps when walking along a straight road, whereas they tend to use AR systems or look at the surrounding scene at turning points. One of the limitations of this research was that too few participants (only three) were recruited for the experiment, which led to poor reliability. In terms of navigation efficiency, Dünser et al. (2012) designed an experiment in which three types of navigation aids, 2D maps, AR, and AR+2D maps, were compared. They found that, compared with 2D maps, it is easier and quicker to navigate to a destination using AR, but more difficult to form a cognitive map. Additionally, AR users saved more cognitive resources, so they were able to perform a second task in parallel during

wayfinding (Wen et al., 2014). However, these studies did not explore the specific cognitive processes of assisted navigation. By contrast, in terms of user experience, Rehr et al. (2012) conducted an experiment to evaluate the navigation efficiency of a voice-based navigation aid, electronic map, and AR for pedestrian users, and found that the electronic map provided a better navigation experience, whereas AR-based navigation often led to poor outcomes. Despite this, it is possible that current users are familiar with electronic maps and inexperienced with AR, which causes poor AR-based performance. Dünser et al. (2012) found that a combination of AR and a 2D map contributed to a higher level of user satisfaction, and a 2D map was better than AR alone in terms of user experience.

To summarize, many studies have evaluated navigation aids, but few have discussed the strengths and weaknesses of AR and 2D maps from the perspective of the human cognitive process. The goal of our study is to understand the cognitive differences between the two aids using a series of experimental tasks. More specifically, we would like to know how the vivid reality provided by AR technology affects a user's wayfinding behavior and performance, visual attention, cognitive processing, spatial memory, and subjective experience. This will then help us to determine the respective advantages of each navigation aid.

2.2. Research method to evaluate the navigation aid users' behavior

Various methods exist for investigating the role of maps in the navigation process. Since Nielsen proposed three aspects of evaluating a map from the perspective of usability engineering – efficiency (response time), effectiveness (accuracy), and satisfaction (Nielsen, 1993) – they have inspired many studies. A conventional method in navigation wayfinding research is to measure the behavioral performance of participants. Accuracy and response time are general metrics that are used to measure the effectiveness and efficiency of map use tasks, respectively (Dong, Yang, et al., 2020). For example, Liao et al. (2016) collected accuracy and response time metrics to identify the difference in user performance between 3D and 2D maps. The higher the accuracy and the lower the response time, the better the user performance.

From the surface of behavior to the deeper level of human spatial cognition, navigation wayfinding contains a series of cognitive processes, which necessitate information intake of visual cues (Afrooz et al., 2014). Eye-tracking technology is a direct method of observing the corresponding visual behavior and studying performance (Ooms et al., 2010). Eye-tracking technology has

helped researchers study cognitive processes in many disciplines. In the field of geographic information science, previous studies integrated eye-tracking methodology and cartographic evaluation (Çöltekin et al., 2009). Subsequently, more researchers adopted this method and began quantitatively evaluating map-using behavior. Keskin et al. (2018) used an eye-tracking technique to record the eye movements of map users when they were learning, acquiring, and memorizing screen map information. The study conducted by Afrooz et al. (2014) also measured participant eye movements to determine which features of landmarks were encoded in memory. Several metrics, such as average fixation duration, average saccade duration, and saccade frequency, are common indices for measuring visual behavior and exploring geographical cognitive processes (Liao et al., 2016). Like previous studies, we used an eye tracker to measure the eye movements of participants in the process of navigation wayfinding.

In addition to studying participants' eye movements from the perspective of visual cognitive processing, each human's geospatial cognition should also be considered and assessed based on the individual's naive feelings. Several post hoc methods have been introduced, and a sketch map is a main experimental method. A sketch map is an effective and innovative method to externalize cognitive maps, including their spatial, temporal, and sequential properties (Huynh & Doherty, 2007). It can be a reliable representation of an internal cognitive map that reflects the process of cognitive mapping (Golledge & Gärling, 2004). Questionnaires, often in the form of Likert-scale items, can transform a subjective experience into quantitative data (Dong, Yang, et al., 2020; Dünser et al., 2012; Wenczel et al., 2017). Interviews often provide more subjective and phenomenological findings and have been applied in previous research (Koletsis et al., 2017). Through questionnaires and interviews, we can not only obtain basic information about participants, but also their feelings and level of satisfaction during the navigation process, and thus we can study their geographical cognitive processing.

In our research, we are interested in the effects of AR aids on visual attention and cognitive processing throughout wayfinding; thus, we adopt eye-tracking technology to delve deeper into the "cognitive" level. Moreover, instead of using a single method, as in previous studies, we comprehensively adopt multiple methods. We recode behavioral metrics from wayfinding videos, and adopt multiple methods, such as eye tracking, sketch maps, and interviews. With both quantitative and qualitative data,

we attempt to explore the geographical cognitive process during and after wayfinding, and thus reveal the distinction between AR navigation and 2D electronic map navigation from multiple angles.

3. Method

3.1. Participants

For this experiment, 73 university students (17 males) with an average age of 19.55 years ($SD = 1.13$ years) were recruited. The participants were mainly from Beijing Normal University and had different academic backgrounds, including geography, psychology, and economics. No participants had professional knowledge or special experience in the domain of navigational systems or AR, but all participants had experience of using smartphone map applications, so they could learn to use AR navigation readily, even if they had never tried it before. None of them had ever walked along the routes in the experiment. Regarding their physical condition, no participants had an eye disease or impaired color vision. The research was approved by the local institutional review board, and each participant received a monetary reward (45 Chinese yuan) for their contribution.

3.2. Apparatus

Eye-tracking glasses (ETG) developed by SMI (Apple Inc.) were adopted to acquire eye-tracking data. It has a sampling rate of 60 Hz, tracking accuracy of 0.5° , and horizontal and vertical tracking ranges of $\pm 80^\circ$ and $\pm 60^\circ$, respectively. The front-facing camera on the glasses can synchronously record the scene video with voice (1280×960 , 24 fps). The ETG was connected to a Lenovo laptop on which recording, calibration, and gaze point visualization were conducted using the iView ETG software platform.

Two portable navigational devices were implemented for the experiment. One was an AR navigation module developed using the Baidu Map mobile app (Baidu, Inc., 2020; <http://map.baidu.com/zt/client/index/>) and a smartphone provided by the experimenter. The AR module overlays a virtual blue track on the video stream recorded by the smartphone camera and participants only need to follow the direction of the track (shown in Figure 1(a)). The other aid was a 2D electronic map developed using AutoNavi Software (Alibaba Group, 2020; <https://mobile.amap.com/>). A blue route line is delineated on the map (shown in Figure 1(b)). We have uploaded recorded videos as Supplementary Materials to illustrate the interface.

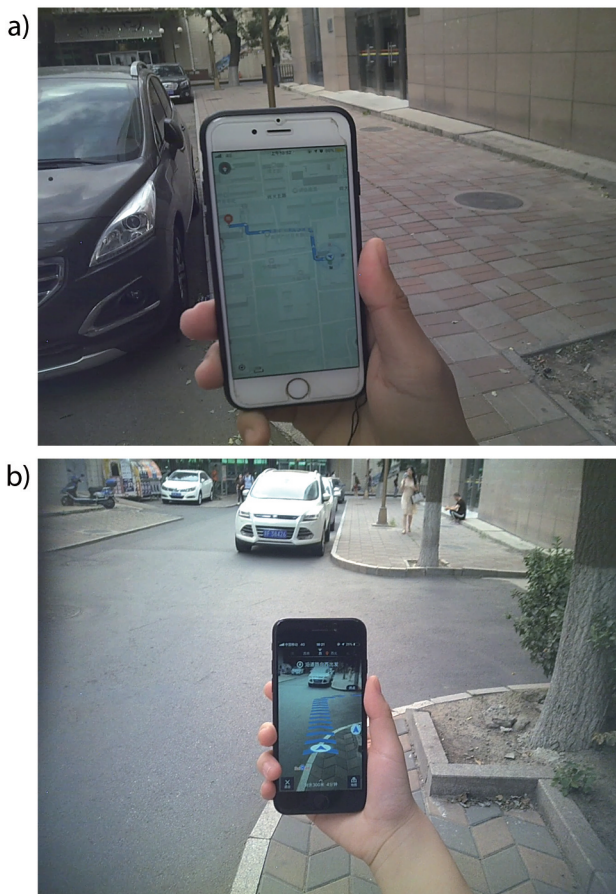


Figure 1. Participants using an AR module (a) and 2D electronic map (b) to navigate themselves. The photos were captured by the eye tracker.

3.3. Procedure

3.3.1. Pre-wayfinding task

All the participants completed the Santa Barbara Sense of Direction Scale (SBSOD; Hegarty et al., 2002) online. The scale contains 15 items. Each item is measured using a 7-point Likert scale (1: “completely disagree” to 7: “completely agree”) on a sentence that describes the participant’s sense of direction. The average score is calculated after reverse scoring, and the higher the score, the better the participant’s large-scale spatial ability. The scores approximately followed a normal distribution ($M = 3.89$, $SD = 1.08$), with a Cronbach’s α of 0.861.

3.3.2. Wayfinding task

The three experimental routes were all located on the campus of Beijing University of Posts and Telecommunications, as shown in Figure 2. The routes were called Route 1 (R1; 255 m), Route 2 (R2; 330 m), and Route 3 (R3; 320 m), and the starting points and destinations were pre-defined. The participants were instructed to find their way

to the destination from the starting point by referring to the mobile navigation app. Each participant performed the wayfinding task on only one of the three routes and used either the AR navigation module or a traditional electronic 2D map navigation system.

The experiment was conducted on both sunny and cloudy days. Each participant donned the ETG with the assistance of the experimenter at the east gate of the campus. Lenses were installed for participants with myopia or hyperopia to correct their vision. Then, the participant walked to the starting point along the pre-defined guiding path under the guidance of the experimenter. The path toward the starting point did not overlap or intersect with the subsequent task route to prevent the participant from becoming familiar with the route in advance.

At the starting point, the experimenter calibrated the ETG using a 3-point calibration method and informed the participant of any warnings and instructions. The participant was told to reach the destination under the assistance of the AR system or 2D electronic map. The participant set off after fully understanding the instructions provided, and the recording process started simultaneously. The experimenter followed the participant, always lagging about 1.0 m behind. As the participant progressed, the experimenter typically did not give any clues about where to go, or answer any questions about the direction and route. A hint was only given if the participant deviated from the correct route for about 10 s and continued to walk in the wrong direction without being conscious of making mistakes, but in fact, only one 2D map participant was given a hint.

Data quality can be significantly influenced by strong light. To boost the performance of the device, the participant wore a sun visor and carried an umbrella for shade, if necessary. When the participant arrived at the destination, the experimenter stopped the recording of the ETG and the participant took it off, which indicated that this phase of the task had ended.

3.3.3. Post-wayfinding task

After finishing the task, the participants were asked to draw their routes onto a paper map of the university campus. The experimenter did not give any indication of the starting point and destination, so the participants had to independently locate themselves and draw their mental maps based on spatial memory.

All participants were then interviewed by the experimenter about the user experience. The interview was structured and participants answered pre-prepared questions (see the Supplementary Materials).

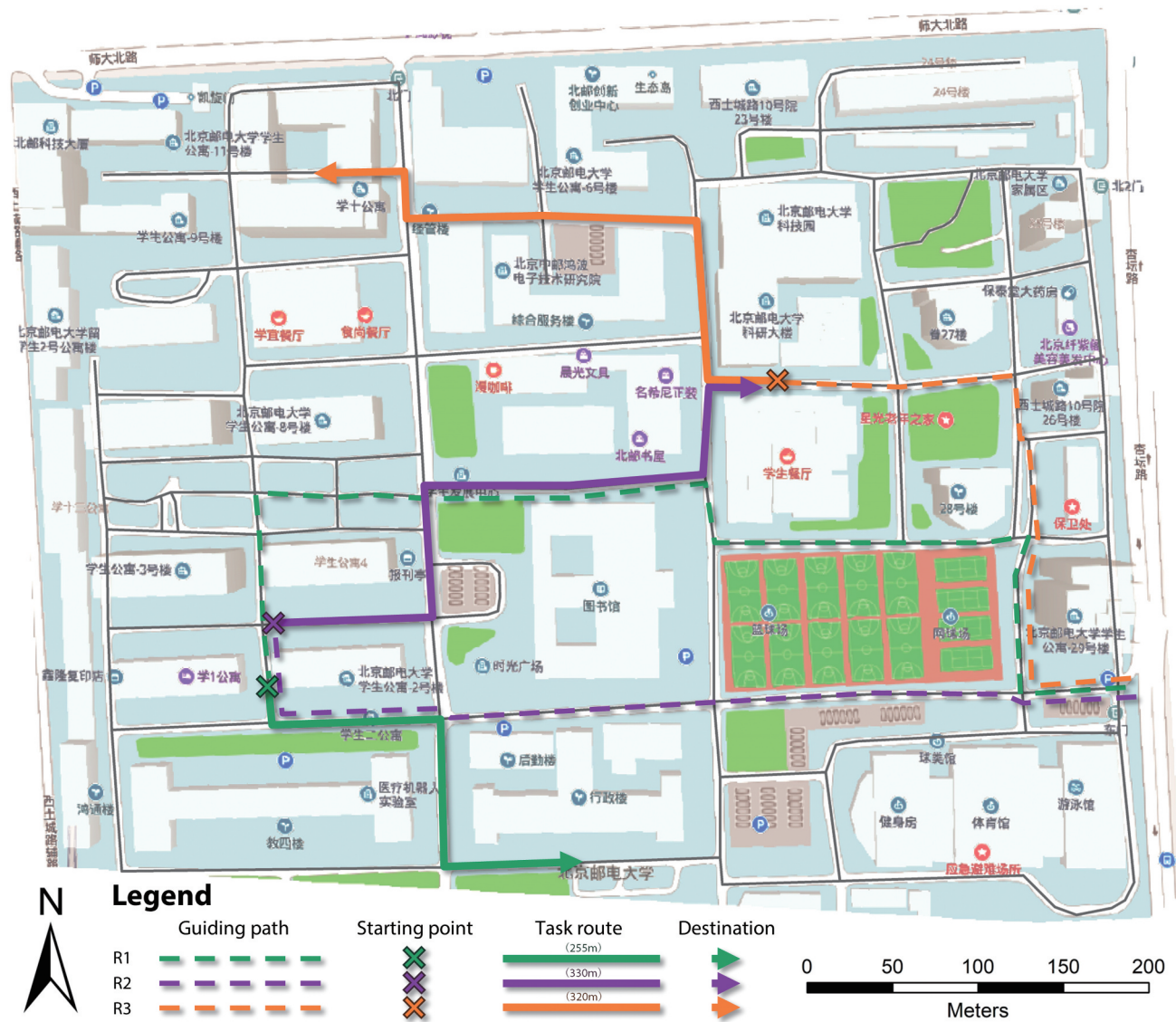


Figure 2. Illustration of the three experimental routes. Dashed lines indicate the path through which participants were guided to the starting points, and solid lines indicate the wayfinding task routes, from the starting points to the destinations. The base map is a screenshot from the Baidu map service (<https://map.baidu.com/>). Permission granted.

3.4. Analysis framework

We obtained four types of data that required further processing and analysis: synchronous video recorded by the front-scene camera, the raw gaze point dataset recorded by the ETG, the sketch route drawn by the participants, and the answers from the qualitative interview. Before we discuss the steps for the data analysis, it is necessary to describe the data preprocessing steps and the critical algorithm adopted in the research to understand the semantic meaning of the participants' visual attention during the wayfinding process.

3.4.1. Data preprocessing

Eye-tracking data sampled by ETG are a time series of the positions of millions of gaze points. We first identified the fixations that were used in subsequent analysis from gaze point series, and then screened the 63 participants whose data quality was relatively high. Next, we adopted Deeplabv3+, which is an advanced deep neural network, to semantically segment each video frame (Chen et al., 2017). The process was implemented by importing the open-source Python library Deeplabv3 into TensorFlow (<https://github.com/tensorflow/models/tree/master/research/deeplab>), and the network was

trained using the Cityscapes dataset (Cordts et al., 2016). Each fixation was overlaid on the corresponding segmented frames and labeled with one of six semantic tags: Building, Person, Ground, Moving Object, Static Object, and Phone. A more detailed description of the steps is provided in the Supplementary Materials.

3.4.2. Wayfinding behavior data

Different navigation aids might cause different behavioral patterns in users, thereby reflecting the levels of efficiency of the aids and human-computer interaction (HCI) rules of the two navigational systems. We obtained several metrics by recoding the scene video manually and keeping records of the frequency of certain behavioral events during the process of wayfinding. Specifically, we used three metrics:

- *Wayfinding duration* (in seconds). This is the duration of the wayfinding process from the starting point to the destination. It reflects the overall efficiency of an aid's navigation performance.
- *Frequency of mistakes or hesitation events* (in number of occurrences per route). This is the frequency of the participant's deviation from the correct route or a hesitating behavioral response in the video. Criteria used to determine a hesitation event were a participant's behavioral response, such as doubt, a gasp, or looking around hesitantly. This metric reflects the low effectiveness or instability of a wayfinding aid when navigating the user: the smaller the value, the better the performance.
- *Frequency of interaction* (in number of occurrences per route). This is defined as the number of times that fixation switches between the device and the wayfinding scene, which reflects the intensity of the HCI. It was calculated based on the annotated fixation data.

For each dependent variable, the route (R1, R2, and R3) and navigation aids (AR and 2D maps) were considered as two factors because the variables could be influenced by both the routes and type of aid. We conducted a two-factor Analysis of Variance (ANOVA) for the three metrics of wayfinding behavior. For ANOVA, two assumptions must be met: 1) the data from each group should represent a normal distribution (the assumption of normality), and 2) the within-group variances of all groups should be identical (the assumption of homogeneity of variances). The two assumptions were tested using the Shapiro–Wilk normality test and Levene's Test using two R packages: *stats* and *car* (<https://cran.r-project.org/web/packages/car/citation.html>) (Fox & Weisberg, 2019), respectively. If the assumptions were

not met, a non-parametric test had to be used. Only one metric, frequency of mistakes or hesitation events, didn't meet the assumptions. Before the statistical analysis was conducted, we standardized all the numeric values (using the Z transformation) within six groups and excluded 3 out of 183 data points whose z-scores were greater than 2 or less than -2.

3.4.3. Eye-tracking data

In traditional eye-tracking research, metrics are divided into two categories: fixations and saccades. We selected seven variables computed from all fixations, commonly adopted in previous research, because of their cogent connection with internal cognitive meaning (Dong et al., 2014; Liao et al., 2018; Liao et al., 2019).

- *Fixation frequency* (in s^{-1}) is calculated by dividing fixation counts by wayfinding duration. It reflects the efficiency of information processing and is typically related to fixation duration (Ooms et al., 2012). Researchers have also claimed higher fixation frequency might indicate semantic importance (Jacob & Karn, 2003).
- *Average fixation duration* (in ms) is the arithmetic average of all fixation durations. The longer the value, the more time users spend decoding, and thus it reflects a deeper cognitive processing level (Just & Carpenter, 1976).
- *Saccade frequency* (in s^{-1}) is calculated by dividing saccade counts by wayfinding duration, and it represents the efficiency of the visual search.
- *Average saccade duration* (in ms) is the arithmetic average of all saccade durations.
- *Average saccade amplitude* (in degrees) is the average amplitude of all saccades, that is, the span of the visual search.
- *Average scan path* (in px/s) is the ratio of total scan path duration to wayfinding duration, which indicates participants' visual search intensity. Combined with saccade measurements, it can indicate whether the participant conducted an extensive visual search during wayfinding (Goldberg & Kotval, 1999).
- *Average pupil diameter* (in mm) reflects the cognitive workload of the navigation process. Research has found that cognitive workload can be reliably reflected by changes in the size of the pupils (Backs & Walrath, 1992; Smerecnik et al., 2010).

As above, we applied a two-factor design for each dependent variable. We considered a two-factor ANOVA first, but none of the six variables followed a normal distribution. This is not surprising because a number of studies have reported that eye-tracking

variables sometimes violate the assumptions of ANOVA (Dupont et al., 2014). Therefore, we adopted a non-parametric test to determine whether any difference in the means existed, which would indicate an effect of the type of aid. Specifically, the Scheirer–Ray–Hare test is a variation of the Kruskal–Wallis H test and is suitable for two-factor non-parametric significance analysis. The confidence α was set to 0.05. All the values of the seven variables were standardized, and 20 outliers (less than 5%) were excluded, which is similar to the method described in Section 3.4.2.

In addition to the global metrics throughout the wayfinding process, we calculated the proportion of fixation durations of each semantic category as follows:

$$\text{visual attention}_k = \frac{\sum_i FD_k}{\sum_i FD},$$

where FD is the duration of a certain fixation and FD_k is the duration of a fixation labeled with certain category (or the k -th semantic category). The ratio of their respective sums represents the relative intensity of visual attention on each semantic category. To check whether the distribution of visual attention among different ground features was influenced by the navigation aid, we conducted an independent-sample T-test between the AR and 2D map data for each semantic category. Similarly, we excluded 14 outliers (less than 4%) in total from both the AR group and 2D group whose standardized z -scores within the group were greater than 2 or less than -2 .

3.4.4. Sketch map data

Sketch maps are the external expression of the internal representation of spatial memory, or a “cognitive map.” By analyzing the data extracted from a sketch map, we can speculate on the cognitive impact on spatial memory caused by the use of the navigation aid. Two indicator variables are considered: whether the sketch route is perfectly correct (I_1) and whether the directions of all the turning points are correct (I_2), as shown in Figure 3. We consider the second variable because we suspect that some participants memorize spatial relationships using egocentric references rather than allocentric references, and thus make spatial decisions by remembering the order of left or right turns.

4. Results

Because the types of navigation aids used by the participants were not randomly defined in advance, it is important to prove that there was no systematic difference in the large-scale spatial ability between the two groups to exclude the possibility that the experimental

effect was caused by individual differences. Hence, we conducted an independent-sample T-test on the SBSOD score between the groups (summarized in Table 1). The results showed that there was no significant systematic difference in the sense of direction between the two navigation aid groups ($t(59.86) = -1.252$, $p = 0.215$). Based on this result, we could continue to analyze other data.

4.1. Wayfinding behavioral data analysis

The frequency of mistakes or hesitation events data did not satisfy the assumptions required for ANOVA. This could have resulted from the low number of mistakes participants made because both the electronic maps and AR system were sufficiently convenient for the participants to navigate. Therefore, we evaluated the difference in the frequency of mistakes or hesitation events using a non-parametric Wilcoxon rank sum test. We conducted ANOVA for the remaining variables (wayfinding duration and frequency of interaction) which passed the assumptions. The behavioral variables are described in Table 2 and the results of ANOVA are shown in Table 3. The test of the frequency of mistakes and hesitation events did not reach a significant level ($W = 423.5$, $p = 0.712$).

Because of the different lengths of the routes, there were unsurprisingly significant differences among them. However, the difference between the two types of aid did not reach a significant level ($p = 0.092 > 0.05$). The interaction effect between the type of aid and route of the wayfinding duration and all between-group effects of the frequency of interaction were not significant.

4.2. Eye-tracking data analysis

The Scheirer–Ray–Hare test was implemented for the seven eye-tracking variables recorded throughout the course of wayfinding. The test results are shown in Table 4 and Figure 4.

The effects of the type of aid were significant for three metrics. For AR participants, their average fixation duration (in ms, $M_{AR} = 237.03$, $SD_{AR} = 29.97$) was significantly shorter than those of 2D participants (in ms, $M_{2D} = 254.37$, $SD_{2D} = 33.19$) and their average saccade amplitude (in degrees, $M_{AR} = 13.05$, $SD_{AR} = 9.31$) was greater than 2D group (in degrees, $M_{2D} = 6.72$, $SD_{2D} = 1.36$). For average pupil diameter, AR participants had smaller

Table 1. Description of the SBSOD score.

Variable	Group	Mean (SD)
SBSOD Score	AR	4.06 (1.34)
	2D	4.39 (0.86)

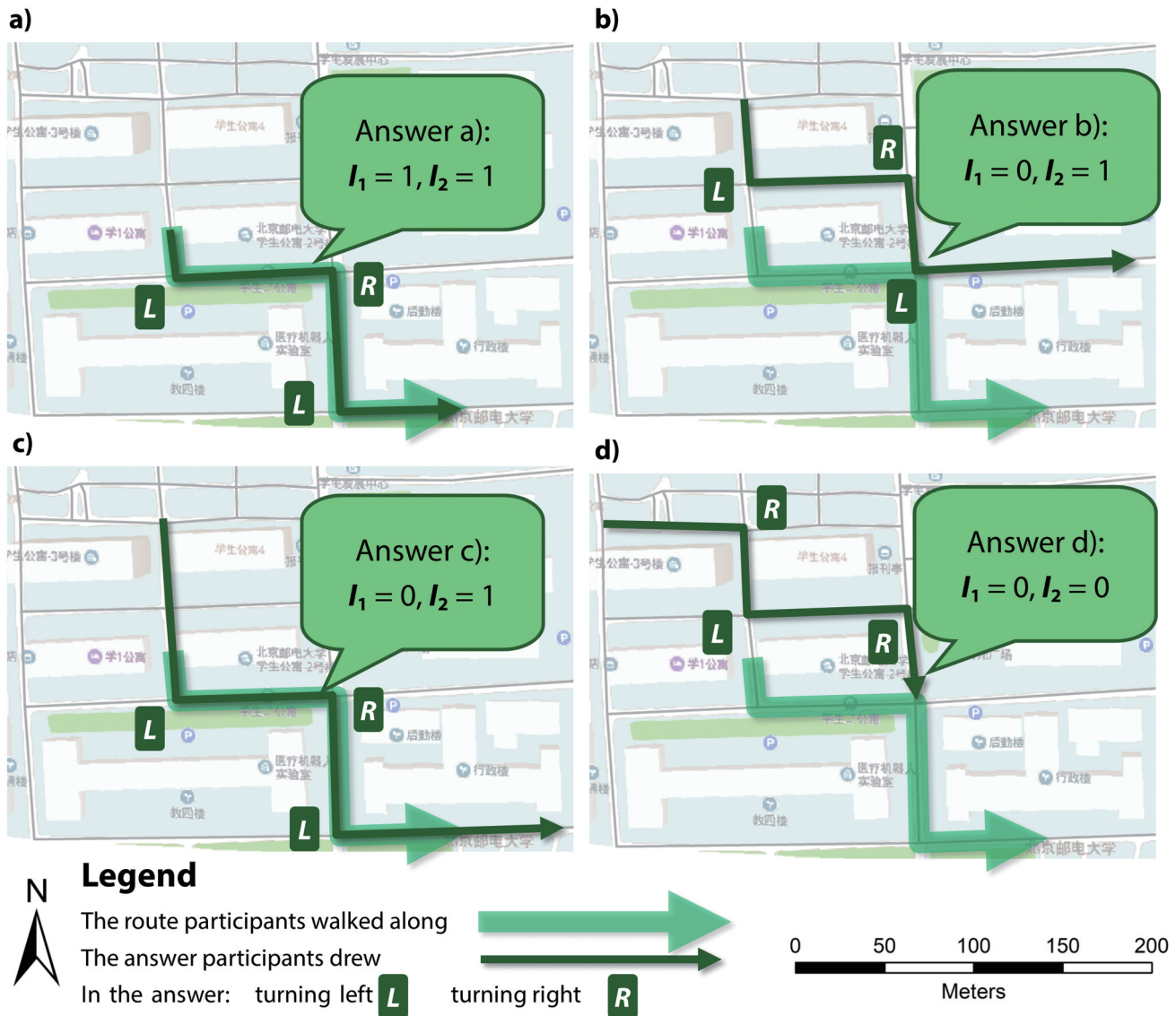


Figure 3. Illustration of the meaning of I_1 and I_2 . (a): Correct answer ($I_1 = I_2 = 1$); (b), (c): For example, a pedestrian will turn left, right, and left if he/she walks along R1. Therefore, sketch answers in figure (b) and (c) have the correct turns (left-right-left, $I_2 = 1$) even if they are not perfectly correct ($I_1 = 0$); (d): The answer doesn't depict correct turns ($I_1 = I_2 = 0$).

Table 2. Description of the behavioral variables.

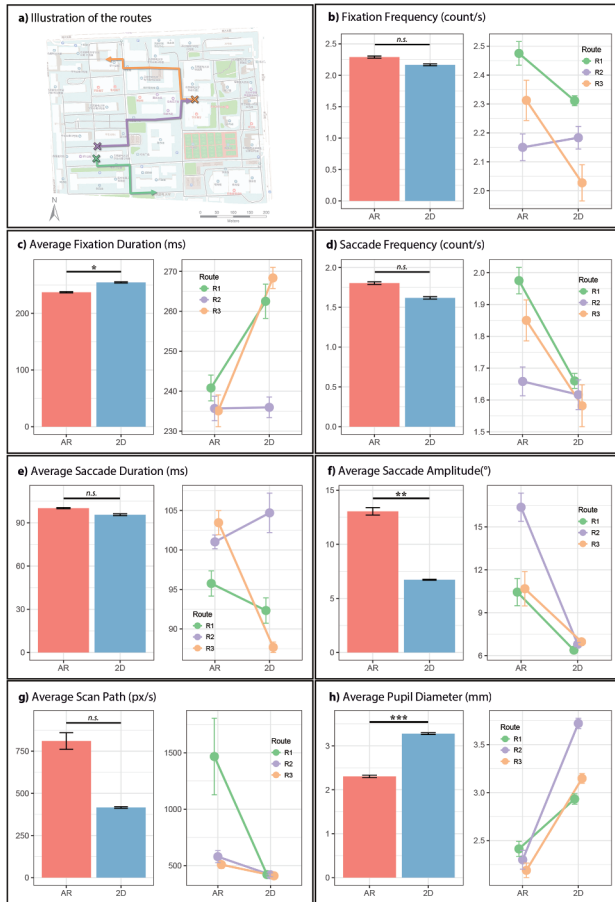
Variable	Group	R1	R2	R3
Wayfinding duration	AR	198.83 (16.87)	248.69 (25.21)	236.93 (22.68)
	2D	196.42 (31.58)	268.44 (38.63)	256.00 (32.49)
Frequency of mistakes or hesitation	AR	1.29 (1.11)	1.50 (1.57)	1.44 (1.74)
	2D	1.45 (1.37)	1.00 (1.55)	1.70 (1.83)
Frequency of interaction	AR	24.56 (11.85)	21.09 (4.89)	26.67 (8.12)
	2D	24.67 (11.54)	21.09 (6.44)	26.11 (7.66)

Table 3. Results of ANOVA on the behavioral variables.

Dependent Variable	$df_{residual}$	Factor	df_{factor}	F	p
Wayfinding duration	58	Type of aid	1	2.929	0.092
		Route	2	25.622	<0.001***
		Interaction	2	0.939	0.397
Frequency of interaction	51	Type of aid	1	0.002	0.963
		Route	2	1.935	0.155
		Interaction	2	0.009	0.991

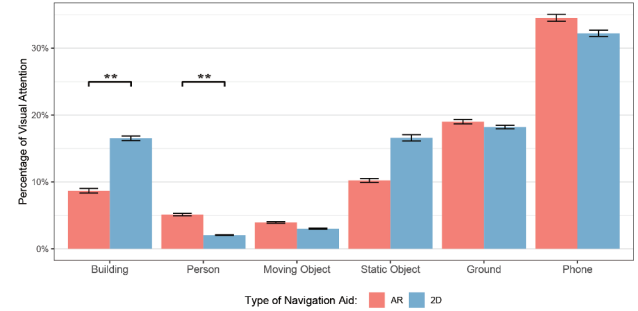
Table 4. Result of ANOVA on the eye-tracking metrics.

Dependent Variable	<i>df</i> _{residual}	Factor	<i>df</i> _{factor}	<i>H</i>	<i>p</i>
Fixation frequency	54	Type of aid	1	2.087	0.149
		Route	2	1.517	0.468
		Interaction	2	1.495	0.474
Average fixation duration	56	Type of aid	1	4.114	0.043*
		Route	2	3.628	0.158
		Interaction	2	2.774	0.250
Saccade frequency	55	Type of aid	1	2.484	0.115
		Route	2	1.777	0.411
		Interaction	2	1.307	0.520
Average saccade duration	55	Type of aid	1	3.545	0.059
		Route	2	3.905	0.142
		Interaction	2	3.667	0.160
Average saccade amplitude	52	Type of aid	1	8.610	0.003**
		Route	2	3.423	0.181
		Interaction	2	1.796	0.407
Average scan path	53	Type of aid	1	2.079	0.149
		Route	2	0.954	0.621
		Interaction	2	1.129	0.569
Average pupil diameter	52	Type of aid	1	20.409	<0.001***
		Route	2	3.659	0.160
		Interaction	2	2.321	0.313

**Figure 4.** Illustration of three routes (a) and comparison of the seven metrics measured for AR and 2D map use with routes ungrouped and grouped (b)-(h). The error bars represent ± 1 standard error. (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$, representing the same meaning hereinafter).**Table 5.** Comparison of the attention of different semantic categories between the groups.

Type of scene object	<i>df</i>	<i>t</i> (AR – 2D)	<i>p</i>
Phone	56	0.411	0.683
Building	56	–3.037	0.004**
Person	36.35 ^a	2.728	0.010**
Ground	56	0.376	0.708
Moving object	47.24 ^a	1.519	0.135
Static object	57	–1.571	0.122

^aThe assumption of homogeneity of the variances was not met, therefore the degrees of freedom were corrected.

**Figure 5.** Percentage of visual attention of different semantic categories in the wayfinding scene for AR and 2D map use. The error bars represent ± 1 standard error.

pupil (in mm, $M_{AR} = 2.30$, $SD_{AR} = 0.81$) than 2D participants (in mm, $M_{2D} = 3.28$, $SD_{2D} = 0.56$). None of the aid type effects were significant for the other variables. The effects of route and interactions were not significant.

To understand the impact of the navigation aids on the semantic meaning of visual attention, we performed a T-test for all seven variables, and the results are shown in Table 5 and Figure 5.

The differences between the two groups for the building and person categories reached a significant level. AR participants allocated less attention on buildings ($M_{AR} = 9.06\%$, $SD_{AR} = 8.87\%$; $M_{2D} = 16.54\%$, $SD_{2D} = 9.82\%$) but more on persons ($M_{AR} = 5.54\%$, $SD_{AR} = 5.21\%$; $M_{2D} = 2.70\%$, $SD_{2D} = 2.07\%$). The differences in the other categories, including the phone category, were not significant.

4.3. Sketch map analysis

We computed the percentage of correct sketch answers within the AR and 2D map groups, as shown in Figure 6. As we mentioned, the analysis distinguished between perfect correctness (I_1) and turning correctness (I_2). Clearly, the accuracy of the sketch maps in the 2D group was much higher than that in the AR group.

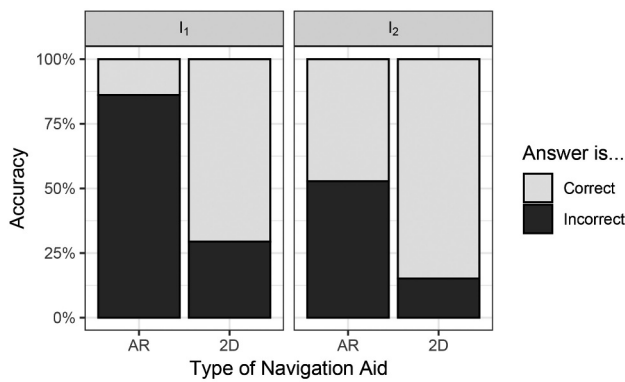


Figure 6. Accuracy of the sketch maps in each group.

Table 6. Results of logistic regression on the sketch responses.

Response	Predictor	β (SE)	p
I_1	SBSOD score	0.75 (0.67)	0.027*
	Type of aid (AR = 0)	2.81 (0.34)	<0.001***
	(intercept)	-5.17 (1.71)	0.002**
I_2	SBSOD score	0.64 (0.28)	0.021*
	Type of aid (AR = 0)	1.70 (0.61)	0.005**
	(intercept)	-2.72 (1.20)	0.023*

To confirm the impact of the AR system on spatial memory and compare it with the impact of a 2D map, we performed logistic regression by setting the scale scores and type of aid characteristic variable (AR is 0; 2D map is 1) as predictors, and I_1 and I_2 as response variables. The results are shown in Table 6.

Both the SBSOD score and type of aid (using 2D instead of AR) significantly positively predicted the correctness of the sketch map, regardless of whether there was perfect correctness or directional correctness.

5. Discussion

5.1. Similar wayfinding performance

AR and 2D maps are two common phone-based navigation aids, and one of the objectives of this experiment was to explore the benefits of using these two aids on wayfinding behavior. We only found a difference in the wayfinding durations among the three routes, which was not surprising because of their different lengths. We did not find any significant differences between the two aids in wayfinding duration, frequency of interaction, and frequency of hesitation and mistakes. Neither did we find an interaction effect between the type of aid and routes. The non-significant results may have several causes. From the interview after the experiment, we know that most of our participants had used the AR aid for the first time (30 participants out of 38), which led to a difference in familiarity between the

groups. This difference could offset the ease of AR compared with the ease of the 2D map. However, the results still demonstrate that 2D electronic maps were not worse than AR in terms of navigation efficiency; that is, both AR and 2D electronic maps provided a favorable navigation service for users, which resulted in a ceiling effect. The AR users' wayfinding times, failure times, and human-machine interactions did not radically differ from those of the 2D electronic map users, both in terms of effectiveness and efficiency.

5.2. AR aids' influence on visual behavior

Concerning visual behavior, the eye-tracking metrics explain the latent cognitive processing during the wayfinding course. The average fixation duration of the 2D map was significantly greater than that of the AR, as was the average pupil diameter. The average fixation duration represents the depth of cognitive processing, or the decoding difficulty of visual information, and the average pupil diameter reflects the cognitive workload of ongoing processing. The difference in the two metrics demonstrates that the AR navigation aid could substantially lower the difficulty of wayfinding. Our interview results also verified that 11 participants expressed that the AR system had "intuitive" characteristics. This finding is consistent with the previous research argument that an AR aid may be easier to use than a 2D map (Wen et al., 2014). Kim and Wohn (2011) stated that instead of using an abstract symbol to convey geoinformation, the AR system presents a real-time scene image flow that is easily recognized, which makes wayfinding decisions easier and decreases cognitive load. For one search measure, that is, average saccade amplitude, the AR result was significantly higher than that of the 2D map. The difference illustrates that the AR users conducted the visual search process across a greater span. We speculate that the lower the level of processing and cognitive workload, the more opportunities users had to search for targets of interest in the surrounding environment.

Video segmentation and annotation revealed some rules about visual attention. We did not find a significant difference in visual attention on the phone screen between the two groups (see Figure 5). This means that the users of the 2D map did not spend significantly more time looking at the phone than the AR users. Although this might sound surprising (because the two aids display spatial information in completely different forms), it reconfirms our finding that the users of both aids had a similar interaction pattern. We propose two possible explanations for this non-significant statistical result: 1) Both types of aid had

a high navigation efficiency. The participants did not have to stare at the screen all the time, and the display style rarely altered the users' attention because the phone screen itself shared quite a small proportion of attention. 2) The two groups of participants gazed at their phones for different reasons. For 2D maps, the participants might have gazed at the screen often when interpreting cartographic symbols, thereby increasing the percentage of phone attention, but for AR, the real-time interface might have been more attractive and interesting than the electronic map, thus attracting more visual attention. The two proposed explanations are both somewhat probable and more exploration is needed in the future.

However, visual attention differed with respect to certain types of ground features, such as buildings and persons. The map users paid more attention to buildings. They might have needed to refer to a building's appearance and structure and compare it with symbols on the 2D map to locate themselves and determine the direction. Instead, the AR users did not rely on buildings to determine the correct path, and we could explain such differences according to the interview results. However, it is surprising that the AR users preferred to look at people walking on the street. We speculate that the AR users, who had lower cognitive workloads, were able to pay more attention to surrounding objects. Pedestrians, having higher visual salience, caught their attention most. Previous research has proved that objects with higher visual salience attract visual attention more easily in the wayfinding scenario (Dong, Qin, et al., 2020). Related psychological studies have also proved that human faces are special and easier to process than other objects (Bruce & Young, 1986; Farah et al., 1995).

5.3. AR aids' influence on spatial memory

Regarding post hoc spatial memory, both large-scale spatial ability (measured by SBSOD) and type of aid affected whether a participant could draw a perfectly correct route or a route with the correct turning points. The higher the participants' environmental spatial ability, the higher their probability of drawing a correct route. The AR users performed far worse at route drawing than the map users. This shows that the ease of using the AR navigation aid blurred the participants' spatial memory, which confirms the argument of previous research (Wen et al., 2014). The interview results (see the Supplementary Materials) did not reveal a difference in distance estimation and direction judgment between the two aids, which indicates that the two types of aid provided assistance for those aspects of spatial

knowledge to similar extent. Many participants indicated that the AR aid lacked spatial knowledge about landmarks, shape of routes, and locations. This feedback is valuable and might explain why the AR participants could not draw a sketch as well as the 2D map participants. However, we should acknowledge the limitation of sketch method that the basemap of sketch route has similar layout to the 2D map, which might make participants using 2D map recall the route much more easily. In addition, other aspects of spatial memory, such as scene recognition, architectural style, and semantic meaning of geographic objects, should be further explored in future work.

The potential impact of navigation aids, including AR, on human navigation ability has long been a concern in academia. Researchers have long worried that a widely used application will give rise to a loss of environmental spatial ability. McKinlay (2016) once speculated that the widespread use of automatic, easy-to-use navigation aids might impair navigation ability, and this supposition is somewhat supported by experimental research (Hejtmánek et al., 2018). As a state-of-the-art and popular navigation aid, AR may also have a similar problem, which was empirically proved by our research. Some studies may inspire us to consider how to prevent, or at least alleviate, the negative effects of AR aids on spatial memory. For instance, Liu and Meng (2020) conducted an indoor AR wayfinding experiment, and suggested adding more semantic icons with text as virtual landmarks to an AR real-time scene interface to enhance users' spatial learning. Lokka and Çöltekin (2017) empirically proved that different levels of realism lead to different effects on route learning. The researchers found that memory performance under moderate levels of realism is significantly better than that for an abstract scene (a low level of realism) or totally real scene (a high level of realism). If this is true, it is possible to influence or even improve route recall accuracy by adjusting the extent of realism of a given scene. Those are inspiring for AR navigation designers and developers.

6. Conclusion and future work

This research combined behavioral measurement, eye-tracking technology, sketch maps, and qualitative interviews to reveal the characteristics of AR navigation and 2D map using behavior. We did not find an explicit distinction in the external wayfinding behavior of users using the two types of aid, but we did find differences in the visual behavior, which reflects distinctive cognitive characteristics. The AR participants tended to have shorter fixation durations, greater saccade amplitudes, and larger pupil sizes than the 2D map

participants. They also preferred to pay more attention to people instead of building landmarks. These results provide evidence that AR may lead to easier spatial cognitive processing and lower cognitive resource costs. This might in turn lead to worse spatial memory performance, which was found through the analysis of sketch maps.

In our primary exploration of this new navigation aids, some participants were trying to use the AR aid for the first time and were not familiar with it, whereas the participants using the 2D map were nearly all experienced users, which meant that the factor of familiarity of the aids was not strictly controlled. Additionally, it should be noted that we could not definitely generalize those points because the Baidu AR navigation system does not represent all types of AR navigation modules. Many participants complained about the application interface design of AR aids, and some even provided advice on how to improve the usability of the AR system, but the design style, elements, and composition of the interface vary largely among different AR navigation aids and may result in users' different behavior and experience. Thus, the shortcomings of the AR system in the research cannot be imputed readily to AR aids as a whole. Statistical results and findings in this study should be carefully interpreted and generalized, and we expect that further research on the topic of AR navigation will reveal its usage mechanisms and provide good advice for AR navigation aid development.

Acknowledgments

We would like to thank Dr. Liao Hua for help with data analysis; Dr. Kimberly Moravec and Dr. Maxine Garcia from Edanz Group (<https://en-author-services.edanzgroup.com/>) for editing a draft of this manuscript; and all the participants in the experiment for their contributions to the study.

Disclosure statement

There are no relevant financial or non-financial competing interests to report.

Funding

This research was supported by the National Key Research and Development Program of China [Grant No. 2017YFB0503602] and National Natural Science Foundation of China [Grant No. 41871366].

ORCID

Weihua Dong  <http://orcid.org/0000-0001-6097-7946>
Yulin Wu  <http://orcid.org/0000-0001-8042-1128>

Tong Qin  <http://orcid.org/0000-0001-5724-2105>
Xinran Bian  <http://orcid.org/0000-0002-5713-2608>
Yan Zhao  <http://orcid.org/0000-0002-9731-7105>
Yanrou He  <http://orcid.org/0000-0002-1355-8332>
Yawei Xu  <http://orcid.org/0000-0002-4100-4142>

Data availability statement

As a result of the protection of the research participants' privacy and subsequent ethical problems, the original eye-tracking data, including video records, cannot be publicly shared. Despite this, we have prepared a tidy data table encoded from the original data, with one participant per line. The data that support the findings of this study are openly available in the Harvard Dataverse at <http://doi.org/10.7910/DVN/LDXSUH>.

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